

Fail-Closed Gradient-to-QEF Contracts for Approximate Implicit CSG

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Abstract

Dual Contouring places one vertex per active cell by minimizing a quadratic error function (QEF) assembled from Hermite points and normals. For an approximate implicit CSG field, however, a small value error does not by itself stabilize the active Boolean branch, the normal, the edge intersection, or the QEF solution. We develop a local, fail-closed contract that connects these quantities. First, a differentiable half-space construction shows that, without a positive branch margin, value errors tending to zero can still cause a fixed normal jump at a hard Boolean switch. A positive result therefore propagates a single-normal contract only on branch-separated cells; unresolved switch cells carry a set-valued normal enclosure or return UNKNOWN. For a binary log-sum-exp smooth minimum, we derive an executable gradient bound that contains an unavoidable coupling from value error to weight and gradient error. We then transfer a field-value budget and directional-derivative lower bound to an edge-root displacement, normalize a gradient contract into a normal bound, and obtain a local Hermite-plane budget. Finally, an observed-side spectral guard and Weyl perturbation yield an absolute displacement bound for a regularized QEF vertex. An executable audit checks 250,000 smooth-Boolean cases, 30,000 nonlinear edge roots, and 60,000 QEF systems with no violation of the complete bounds. The QEF audit includes 20,000 non-concurrent systems: inconsistent analytic planes and Hermite patches sampled from spheres and paraboloids. Omitting the smooth value–gradient coupling causes 31,479 failures; omitting the Hermite point budget causes 2,874 QEF-bound failures across the two QEF audits. In the original 40,000-system conditioning audit, regularization makes every bound finite, while a geometry-only identifiability guard is positive in 50.99%. Only 17.42% meet a strict one-percent-of-cell tolerance, exposing the conservatism of a worst-case certificate. At a tolerance of 0.10 cell units, deployable coverage is 46.75%; replacing declared budgets by realized errors raises the diagnostic coverage to 53.52%, while using exact matrix perturbations raises it to 76.32%. Thus both input-contract width and analytical aggregation contribute to the slack. At $\tau = 0.10$, measured displacement is within tolerance in 99.8475% of those 40,000 systems, showing that the observed non-certification gap is almost entirely certificate slack. On a frozen corpus of 528 STEP files, OpenCascade imports and validates every solid subset; a no-healing mesh gate admits 511 models. A standalone regular-grid Dual Contouring pipeline completes 118 of 122 selected models and audits 162,216 active QEF cells with zero complete-bound violations. Its 22.73% coverage is explicitly oracle-diagnostic because the CAD files do not provide certified upstream value and gradient budgets. The result is a verifiable cell-level interface for future certified contouring, not a claim that arbitrary sampled or neural fields already provide the required upstream budgets.

1 Introduction

Signed distance fields (SDFs) support compact Boolean modeling and simple surface interrogation. Their numerical approximations are commonly consumed by contouring algorithms as if the

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value, gradient, projected surface point, and local tangent plane were mutually consistent. This assumption is fragile after approximation and CSG composition. A hard minimum can switch its active operand under an arbitrarily small value perturbation. A smooth minimum avoids non-differentiability, but the interpolation weights depend on the uncertain values and therefore transfer value error into gradient error. Even a bounded normal error does not bound a QEF vertex unless the Hermite point locations and the linear-system conditioning are controlled as well.

The original Dual Contouring method assumes Hermite data consisting of exact intersections and normals [1]. It recognizes that noisy, nearly-coplanar normals make the QEF nearly singular and uses experimentally chosen singular-value truncation. Later work models point and normal noise probabilistically [2], or avoids arbitrary gradient access when reconstructing from sampled SDF data [11]. Recent SDF contouring also estimates gradients and filters unreliable contact points [12]. These methods address important reconstruction and robustness questions, but they do not provide the worst-case compositional contract studied here: given explicit upstream value and gradient budgets for a CSG cell, decide whether a particular Hermite/QEF query has a checkable error bound.

Several neighboring lines of work certify different objects. Distance-estimate analysis tracks value or tracing precision through CSG operations [8]; Lipschitz pruning identifies a regionally equivalent subtree without propagating derivative uncertainty [9]; and differentiable Boolean operators are designed for continuous optimization rather than worst-case certification [10]. Producer-side work such as HotSpot improves the distance semantics of learned fields [7]. These results strengthen the motivation for explicit contracts, but they do not by themselves transfer bounded value and gradient uncertainty to a per-cell Hermite-plane and regularized-QEF displacement bound.

We make six contributions.

1. We prove that hard Boolean fields have no uniform, branch-independent single-normal continuity near a switch, even when the input gradient errors are zero and the value errors tend to zero.
2. We replace a nonexistent global single-normal rule by two sound cases: branch-separated inheritance and a set-valued normal enclosure on unresolved switch cells.
3. We derive a binary smooth-min gradient bound with a computable value–gradient coupling term.
4. We prove a contract chain from an edge-root budget through a normalized normal and local Hermite plane to an observed-side regularized QEF bound.
5. We provide a deterministic, frozen audit with two falsifying ablations, and report both soundness evidence and certificate conservatism.
6. We quantify where the certificate is useful rather than reporting soundness alone. At tolerance $\tau = 0.10$, deployable coverage ranges from 98.31% on orthogonal systems to 9.33% on the 20° edge systems. A separate 20,000-case residual-stress audit covers non-concurrent analytic planes and sphere/paraboloid Hermite patches.
7. We execute a frozen 528-model STEP/B-rep preflight and a 122-model regular-grid Dual Contouring audit. The experiment preserves every rejection in the denominator, cross-checks OpenCascade through a second frontend and version, and separates measured oracle diagnostics from deployable contracts.

Our scope is intentionally local. We do not claim a certificate producer for arbitrary neural fields, a crack-free adaptive octree theorem, or a complete composition theorem for deep CSG trees mixing union, intersection, and difference. The real-CAD implementation is a uniform-grid experimental pipeline, not a production adaptive contourer.

2 Related work

Hermite contouring and QEFs. Extended Marching Cubes and Dual Contouring use surface intersections and normals to preserve sharp features. Ju et al. [1] express a cell’s QEF as

$$E(x) = \sum_i (n_i^\top (x - p_i))^2 = \|Nx - h\|_2^2,$$

and discuss QR representation and SVD truncation for numerical robustness. Their thresholding rule handles rank deficiency in practice but does not certify the output under bounded perturbations of the Hermite inputs. Probabilistic Quadrics [2] instead minimize an expected error under Gaussian point and normal distributions. That work is a principled uncertainty model and prevents us from describing all previous QEF treatment as heuristic. Our distinction is logical and statistical: we propagate deterministic worst-case sets and return UNKNOWN when their worst case cannot be certified; probabilistic quadrics optimize an expected quadratic error under a chosen distribution.

Topology-oriented extensions certify complementary properties. Manifold Dual Contouring guarantees manifold output under adaptive simplification [3], while interval subdivision can produce isotopic approximations of sufficiently regular implicit surfaces [4]. Such connectivity and isotopy guarantees do not bound how errors in an approximate field move a particular Hermite point, normal, or QEF minimizer. Conversely, the cell-level certificate developed here does not imply a crack-free, manifold, or isotopic global mesh.

CSG values, activity, and differentiability. Bálint et al. [8] analyze signed-distance estimates under CSG operations and quantify region-dependent precision loss. Lipschitz Pruning [9] conservatively removes primitives that cannot affect a spatial region and supports hard and smooth CSG. Liu et al. [10] instead make the Boolean operator differentiable with respect to its type for inverse modeling. These works respectively address distance-value semantics, regional expression equivalence, and optimizable operator design. The present question is different: conditional on explicit value and gradient budgets, what can a downstream Hermite/QEF consumer certify for one query?

Approximate and learned fields. Neural Dual Contouring predicts vertices and topology directly from sampled fields [5]. DualMesh-UDF observes that neural UDF values and gradients are unreliable near the zero set and cut locus, and estimates tangent planes from filtered off-surface samples [6]. Carrera et al. [11] reconstruct sharp features from discrete SDF samples without requiring arbitrary gradient queries. Kohlbrenner and Alexa [12] approximate gradients from a tessellation and filter infeasible contact points before surface reconstruction. These methods demonstrate that gradient acquisition is itself a substantive problem. The present contract consumes a certified gradient enclosure; it does not claim to manufacture one for these inputs. HotSpot [7] gives an asymptotically sufficient training condition for convergence toward a true neural SDF. That producer-side guarantee is stronger than an Eikonal loss alone, but it is not a finite, per-query value–gradient budget of the form consumed here.

Nonsmooth and verified geometry. At a hard Boolean switch, the classical gradient need not exist. We use the Clarke generalized gradient [13] only as a safe set-valued enclosure, not as a unique surface normal. Interval and isotopy-preserving implicit meshing methods [4] show the value of fail-closed local predicates, but our target is specifically the transfer of field uncertainty to Hermite planes and QEF vertices.

3 Contracts and semantics

We use the convention that an SDF is positive outside. A hard union is therefore $\min(\phi_A, \phi_B)$, and a hard intersection is the corresponding maximum.

Definition 1 (Local gradient field contract). For a cell $C \subset \mathbb{R}^d$, a contract for an approximate field $\hat{\phi}$ records a value-error budget ε , a set-valued gradient enclosure G_C , the value and gradient validity domains, and certified branch information. In the ball representation,

$$|\hat{\phi} - \phi| \leq \varepsilon, \quad \|\nabla \hat{\phi} - \nabla \phi\| \leq g \quad \text{on } C.$$

Every consumer returns one of three logical states. CERTIFIED means all preconditions hold and an explicit bound is returned. UNKNOWN means the geometry may be valid but the available information is insufficient. INVALID means a validity domain or basic input contract was violated. A heuristic centroid or SVD fallback after UNKNOWN is an *uncertified fallback*; it is not silently reclassified as safe.

4 Hard Boolean gradients

The first result is deliberately stated at differentiability points, avoiding the ill-posed comparison of classical gradients directly on a hard-min tie.

Theorem 2 (No branch-independent single-normal continuity). *Fix $\theta \in (0, \pi]$. In every neighborhood of the origin there are exact half-space SDFs ϕ_A, ϕ_B , approximate SDFs $\hat{\phi}_A, \hat{\phi}_B$, and points x_ε at which both hard minima are differentiable, such that*

$$\|\hat{\phi}_i - \phi_i\|_\infty = \varepsilon, \quad \nabla \hat{\phi}_i = \nabla \phi_i,$$

but

$$\|\nabla \min(\hat{\phi}_A, \hat{\phi}_B)(x_\varepsilon) - \nabla \min(\phi_A, \phi_B)(x_\varepsilon)\| = 2 \sin(\theta/2).$$

Proof. Choose unit vectors n_A, n_B with angle θ , set $d = n_B - n_A$, and let $\phi_i(x) = n_i^\top x$. For $\varepsilon > 0$, define

$$x_\varepsilon = \frac{\varepsilon}{\|d\|^2} d, \quad \hat{\phi}_A = \phi_A + \varepsilon, \quad \hat{\phi}_B = \phi_B - \varepsilon.$$

Then $\phi_B(x_\varepsilon) - \phi_A(x_\varepsilon) = \varepsilon > 0$, so the exact minimum uniquely selects A . The approximate gap is $\hat{\phi}_B - \hat{\phi}_A = -\varepsilon < 0$, so the approximate minimum uniquely selects B . The two gradients are therefore n_A and n_B , whose distance is $\sqrt{2 - 2 \cos \theta} = 2 \sin(\theta/2)$. Finally $x_\varepsilon \rightarrow 0$, while every x_ε remains off the tie. $\square$

The result does not say that hard Boolean gradient information is impossible. It identifies the missing precondition.

Theorem 3 (Branch-separated inheritance). *Suppose $|\hat{\phi}_i - \phi_i| \leq \varepsilon_i$ throughout C , and*

$$\sup_C \hat{\phi}_A + \varepsilon_A < \inf_C \hat{\phi}_B - \varepsilon_B. \tag{1}$$

Then $\phi_A < \phi_B$ throughout C . Hence the hard union equals ϕ_A on C , and every valid local value or gradient contract for branch A transfers unchanged.

Proof. For any $x \in C$,

$$\phi_A(x) \leq \hat{\phi}_A(x) + \varepsilon_A \leq \sup_C \hat{\phi}_A + \varepsilon_A,$$

whereas

$$\phi_B(x) \geq \hat{\phi}_B(x) - \varepsilon_B \geq \inf_C \hat{\phi}_B - \varepsilon_B.$$

Equation (1) gives the strict ordering. $\square$

Proposition 4 (Set-valued switch enclosure). *If ϕ_A, ϕ_B are continuously differentiable near a tie x , then the Clarke generalized gradient of $\psi = \min(\phi_A, \phi_B)$ obeys*

$$\partial_C \psi(x) \subseteq \text{conv}\{\nabla \phi_A(x), \nabla \phi_B(x)\}.$$

For operand enclosures G_A, G_B , the convex hull $\text{conv}(G_A \cup G_B)$ is therefore a safe cell-level enclosure.

Proof. Since ϕ_A and ϕ_B are continuously differentiable near x , each is Clarke regular there, with $\partial_C \phi_i(x) = \{\nabla \phi_i(x)\}$. For a finite pointwise minimum, the Clarke generalized gradient is contained in the convex hull of the generalized gradients of the active operands; at a tie both operands are active. The first inclusion follows. Monotonicity of the convex hull under set inclusion then gives the enclosure $\text{conv}(G_A \cup G_B)$. $\square$

This enclosure is not a unique normal. Near a sharp Boolean feature, keeping multiple constraints may be preferable to averaging them, because the QEF uses their directional diversity to locate an edge or corner.

5 Smooth Boolean gradient propagation

For $k > 0$, define

$$\psi_k(a, b) = -k \log(e^{-a/k} + e^{-b/k}), \quad w_A = \frac{e^{-a/k}}{e^{-a/k} + e^{-b/k}}, \quad w_B = 1 - w_A.$$

Then $\nabla \psi_k = w_A \nabla \phi_A + w_B \nabla \phi_B$.

Theorem 5 (Executable binary smooth-min bound). *At a point x , assume*

$$|\hat{\phi}_i - \phi_i| \leq \varepsilon_i, \quad \|\nabla \hat{\phi}_i - \nabla \phi_i\| \leq g_i.$$

Let $\hat{w}_i$ be the weights computed from the approximate values. If

$$M_{AB} \geq \|\nabla \phi_A - \nabla \phi_B\|,$$

then

$$\|\nabla \hat{\psi}_k - \nabla \psi_k\| \leq \hat{w}_A g_A + \hat{w}_B g_B + \frac{\varepsilon_A + \varepsilon_B}{4k} M_{AB}. \quad (2)$$

The executable choice

$$M_{AB} = \|\nabla \hat{\phi}_A - \nabla \hat{\phi}_B\| + g_A + g_B \quad (3)$$

depends only on approximate gradients and their input budgets.

Proof. Adding and subtracting the true gradients under the approximate weights gives

$$\begin{aligned} \nabla \hat{\psi}_k - \nabla \psi_k &= \hat{w}_A (\nabla \hat{\phi}_A - \nabla \phi_A) + \hat{w}_B (\nabla \hat{\phi}_B - \nabla \phi_B) \\ &\quad + (\hat{w}_A - w_A) (\nabla \phi_A - \nabla \phi_B). \end{aligned}$$

The first line is bounded by $\hat{w}_A g_A + \hat{w}_B g_B$. Writing $w_A = \sigma((b-a)/k)$, the logistic derivative satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 1/4$. The mean-value theorem yields

$$|\hat{w}_A - w_A| \leq \frac{\varepsilon_A + \varepsilon_B}{4k}.$$

The triangle inequality gives (2), and a second triangle inequality gives (3). $\square$

Remark 6 (No unit-gradient assumption). The executable bound does not assume $\|\nabla\phi_A\| = \|\nabla\phi_B\| = 1$. Its proof uses only the stated value and gradient budgets. In particular, the choice in (3) remains valid for non-unit true gradients because it follows directly from the triangle inequality.

The $(1/k)$ factor is not an artifact: the operator becomes ill-conditioned as it approaches a hard switch. The audit in section 8 explicitly tests the tempting but unsound rule obtained by deleting the second term of (2).

6 Hermite contracts

We next bridge field uncertainty to the plane constraints actually consumed by a QEF.

Theorem 7 (Certified edge-root displacement). *Parameterize an edge by arc length $s \in [0, L]$. Suppose $f \in C^1$ has a root $s_\star$ and satisfies*

$$|f'(s)| \geq m_e > 0 \quad \text{on } [0, L].$$

Let $\|\hat{f} - f\|_\infty \leq \varepsilon_\phi$, and let $\hat{s} \in [0, L]$ satisfy $|\hat{f}(\hat{s})| \leq \eta_{\text{solve}}$. Then f is strictly monotone, its root is unique, and

$$|\hat{s} - s_\star| \leq \eta_p := \frac{\varepsilon_\phi + \eta_{\text{solve}}}{m_e}. \quad (4)$$

Proof. Continuity and the positive lower bound prevent f' from changing sign. Moreover $|f(\hat{s})| \leq \varepsilon_\phi + \eta_{\text{solve}}$. The mean-value theorem between $s_\star$ and $\hat{s}$ gives (4). $\square$

A sign change alone does not imply uniqueness; the directional-derivative lower bound is a real contract obligation.

Lemma 8 (Normalized direction). *If $\|u\| \geq \mu > 0$ and $\|u - v\| \leq g < \mu$, then*

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \leq \frac{\sqrt{2}g}{\mu} =: \eta_m. \quad (5)$$

Proof. Let $\alpha \in [0, \pi]$ be the angle between u and v , so that $\|u/\|u\| - v/\|v\|\| = 2\sin(\alpha/2)$. The point v lies in the closed ball of radius g about u . Since $g < \|u\|$, this ball excludes the origin, and its angular radius as seen from the origin is $\arcsin(g/\|u\|)$. Hence $\alpha \leq \arcsin t$, where $t := g/\|u\| \leq g/\mu < 1$. The half-angle identity gives

$$2\sin(\tfrac{1}{2}\arcsin t) = \sqrt{2(1 - \sqrt{1 - t^2})}.$$

The ratio

$$F(t) := \frac{\sqrt{2(1 - \sqrt{1 - t^2})}}{t}$$

increases from 1 to $\sqrt{2}$ on $(0, 1)$. Therefore $\|u/\|u\| - v/\|v\|\| \leq \sqrt{2}t \leq \sqrt{2}g/\mu$. $\square$

Let c be the cell center, and write a local Hermite plane as $n^\top y = h$, where $y = x - c$ and $h = n^\top(p - c)$.

Lemma 9 (Local plane offset). *If*

$$\|p - c\| \leq R_c, \quad \|\hat{p} - p\| \leq \eta_p, \quad \|\hat{n} - n\| \leq \eta_n,$$

and both normals are unit length, then

$$|\hat{h} - h| \leq \eta_p + R_c\eta_n =: \eta_h^{\text{loc}}. \quad (6)$$

Proof. Expand

$$\hat{h} - h = (\hat{n} - n)^\top (p - c) + \hat{n}^\top (\hat{p} - p)$$

and apply Cauchy–Schwarz. $\square$

Centering the coordinate system at the cell makes the bound invariant under a global translation and keeps it proportional to cell scale.

7 Observed-side regularized QEF certificate

Stack m unit normals as rows of $N \in \mathbb{R}^{m \times d}$ and the local plane offsets into $h \in \mathbb{R}^m$. We analyze

$$\min_y \|Ny - h\|_2^2 + \lambda \|y\|_2^2, \quad H = N^\top N + \lambda I, \quad q = N^\top h. \quad (7)$$

The regularization provides a controlled fallback from exact feature intersection to a strongly convex cell objective. Unregularized rank-truncated QEFs require a separate certified rank-selection argument.

Suppose the Hermite contracts give row-wise budgets

$$\|n_i - \hat{n}_i\| \leq \eta_{n,i}, \quad |h_i - \hat{h}_i| \leq \eta_{h,i},$$

and set

$$\eta_N = \left(\sum_i \eta_{n,i}^2 \right)^{1/2}, \quad E_h = \left(\sum_i \eta_{h,i}^2 \right)^{1/2}. \quad (8)$$

Lemma 10 (Observed perturbation budgets). *The quantities*

$$\eta_H = 2 \left\| \hat{N} \right\|_2 \eta_N + \eta_N^2, \quad (9)$$

$$\eta_q = \left\| \hat{N} \right\|_2 E_h + \eta_N \left\| \hat{h} \right\|_2 + \eta_N E_h \quad (10)$$

satisfy $\left\| H - \hat{H} \right\|_2 \leq \eta_H$ and $\|q - \hat{q}\|_2 \leq \eta_q$.

Proof. Let $E = N - \hat{N}$ and $e = h - \hat{h}$. Equation (8) gives $\|E\|_2 \leq \|E\|_F \leq \eta_N$ and $\|e\|_2 \leq E_h$. Expand

$$H - \hat{H} = \hat{N}^\top E + E^\top \hat{N} + E^\top E$$

and

$$q - \hat{q} = \hat{N}^\top e + E^\top \hat{h} + E^\top e.$$

Submultiplicativity gives (9)–(10). $\square$

Remark 11 (Structured row certificates). The aggregation in (8) is deliberately worst case: $\|E\|_2 \leq \|E\|_F \leq \eta_N$. A smaller realized value of $\left\| N - \hat{N} \right\|_2$ is not deployable because the true matrix N is unknown. If an upstream producer can instead certify a computable operator-norm budget $\eta_N^{\text{op}} \geq \|E\|_2$, then lemma 10 and theorem 12 remain valid after replacing η_N by η_N^{op} in (9)–(10). Thus exploiting cross-row structure requires an explicit upstream certificate; an empirical oracle ratio is not a runtime guard.

Theorem 12 (Certified regularized QEF displacement). *Let $\lambda \geq 0$, and let $\hat{y}$ solve $\hat{H}\hat{y} = \hat{q}$, and define the computable lower bound*

$$\underline{\gamma} := \max\{\lambda, \lambda_{\min}(\hat{H}) - \eta_H\}. \quad (11)$$

If $\underline{\gamma} > 0$, then H is positive definite and the true solution y satisfies

$$\boxed{\|\hat{y} - y\|_2 \leq \frac{\eta_H \|\hat{y}\|_2 + \eta_q}{\underline{\gamma}}}. \quad (12)$$

Proof. Because $H = N^\top N + \lambda I$, we have $\lambda_{\min}(H) \geq \lambda$. Weyl's inequality [14] and lemma 10 also give

$$\lambda_{\min}(H) \geq \lambda_{\min}(\hat{H}) - \|H - \hat{H}\|_2 \geq \lambda_{\min}(\hat{H}) - \eta_H.$$

Taking the larger lower bound gives $\lambda_{\min}(H) \geq \underline{\gamma} > 0$, and hence $\|H^{-1}\|_2 \leq 1/\underline{\gamma}$. Using $Hy = q$ and $\hat{H}\hat{y} = \hat{q}$,

$$H(\hat{y} - y) = (H - \hat{H})\hat{y} + (\hat{q} - q).$$

Taking norms proves (12). $\square$

Remark 13 (Well-posedness versus useful certification). For $\lambda > 0$, $\underline{\gamma} \geq \lambda > 0$, so the QEF well-posedness test in (11) alone does not produce UNKNOWN. This does not make the full consumer unconditional: validity domains, Boolean branch state, edge-root and normal preconditions must still pass, and the vertex is CERTIFIED at tolerance τ only when the right-hand side of (12) is at most τ . A finite but larger bound produces UNKNOWN; any centroid or other heuristic used afterwards remains an explicitly uncertified fallback.

The lower bound (11) and displacement bound (12) depend on the observed approximate matrix and upstream budgets, not on an unknown true condition number. For $\lambda > 0$, (11) is always positive; if the data term is nearly singular, it falls back to λ and the resulting displacement bound may be very loose. For $\lambda = 0$, a positive observed spectral gap is required. A consumer requiring tolerance τ accepts the vertex only when the right-hand side of (12) is at most τ .

Remark 14 (Why a direct Neumann substitution does not tighten the guard). Suppose first that $\hat{H}$ is positive definite and $\|\hat{H}^{-1}\|_2 \eta_H < 1$, equivalently $\eta_H < \lambda_{\min}(\hat{H})$. The standard Neumann-series estimate then gives

$$\|H^{-1}\|_2 \leq \frac{\|\hat{H}^{-1}\|_2}{1 - \|\hat{H}^{-1}\|_2 \eta_H} = \frac{1}{\lambda_{\min}(\hat{H}) - \eta_H}.$$

This is exactly the inverse of the observed spectral term already present in (11). Because $\underline{\gamma}$ is the maximum of that term and λ , we have

$$\frac{1}{\underline{\gamma}} \leq \frac{1}{\lambda_{\min}(\hat{H}) - \eta_H}.$$

Thus a direct Neumann substitution, with the same perturbation budgets and numerator, cannot tighten (12). When the Neumann condition fails, it provides no finite inverse estimate, whereas $\lambda > 0$ still yields $\underline{\gamma} \geq \lambda$. A tighter certificate must therefore exploit additional structure in the row perturbations or reduce the upstream budgets, rather than merely replacing the spectral argument. The computable operator-norm alternative in remark 11 states what additional information would be sufficient.

Runtime rule. For each cell, the consumer performs the following steps.

1. Check value/gradient validity domains and the Boolean branch state.
2. Certify each edge root, normalized normal, and local plane offset.
3. Assemble $\hat{N}, \hat{h}$ and budgets (8)–(10).
4. If $\lambda = 0$ and (11) is not positive, return UNKNOWN.
5. Otherwise solve the QEF and evaluate (12). Return CERTIFIED at tolerance τ only if the bound is at most τ ; otherwise return UNKNOWN together with the finite bound. Any downstream fallback remains uncertified.

Table 1: Frozen executable audit. “Ablation failures” count observed violations after deleting a required proof-chain term.

Experiment	tested	complete-bound cases	complete failures	ablation failures
Hard switch construction	30	30	0	–
Smooth-min propagation	250,000	250,000	0	31,479
Hermite edge root	30,000	30,000	0	–
Regularized QEF	40,000	40,000	0	2,097
Non-concurrent QEF stress	20,000	20,000	0	777

8 Executable audit

8.1 Protocol

The frozen experiment uses NumPy, seed 20260808, and analytic fields or finite-dimensional QEF systems. The figure script reads the resulting JSON and recomputes no experiment. Every complete-bound experiment counts violations independently. Two ablations are required to fail:

1. delete the value–gradient coupling term from theorem 5;
2. propagate normal error to the QEF but omit the root-position contribution η_p from the plane-offset budget (6).

The hard-switch test uses half-space angles $30^\circ, 90^\circ, 150^\circ$ and value budgets from 10^{-1} to 10^{-10} . The smooth-min test samples values, $k \in [0.01, 0.5]$, value budgets, gradient budgets, and non-unit true gradient norms. The Hermite test uses monotone cubic fields perturbed by a bounded sinusoid and solves a bracketed approximate root. The first QEF audit uses six compatible planes sharing a common point in four normal configurations: orthogonal, random, a 20° edge, and near-planar. This isolates conditioning but gives the unregularized data term a zero-residual solution. A second audit removes that restriction. It contains 10,000 plane systems with a controlled offset component outside the range of the normal matrix, plus 5,000 sphere and 5,000 paraboloid Hermite patches. Unit-cube local coordinates, varying regularization, normal budgets, and point budgets are used throughout.

Two nondeployable diagnostic arms separate sources of QEF slack. The first replaces declared row budgets by the normal and point errors that actually occurred in each synthetic trial, while retaining the same proof formulas. The second uses the exact matrix and right-hand-side perturbations together with the true smallest eigenvalue. Both arms see true quantities unavailable to a runtime consumer; they diagnose conservatism and are not additional certificates.

8.2 Results

Table 1 reports the principal counts. The complete smooth-min and Hermite bounds show no observed violation. The one-sided 95% exact lower bounds on their finite-sample success probabilities are 99.9988% and 99.9900%, respectively. These are statistical descriptions of the audit, not replacements for the proofs.

For QEFs, all 40,000 regularized systems have a finite bound and show zero observed violations; the one-sided 95% lower confidence bound on this finite success proportion is 99.9925%. In 20,396 systems (50.99%), the observed data matrix itself has a certified positive spectral margin after subtracting the perturbation budget. This geometry-only identifiability is highly configuration dependent (table 2): it holds for almost all orthogonal and random systems, for none of the exact two-normal edge systems, and for only 4.85% of near-planar systems. The regularizer still supplies a finite lower bound in the remaining cases, but the certificate is then usually loose.

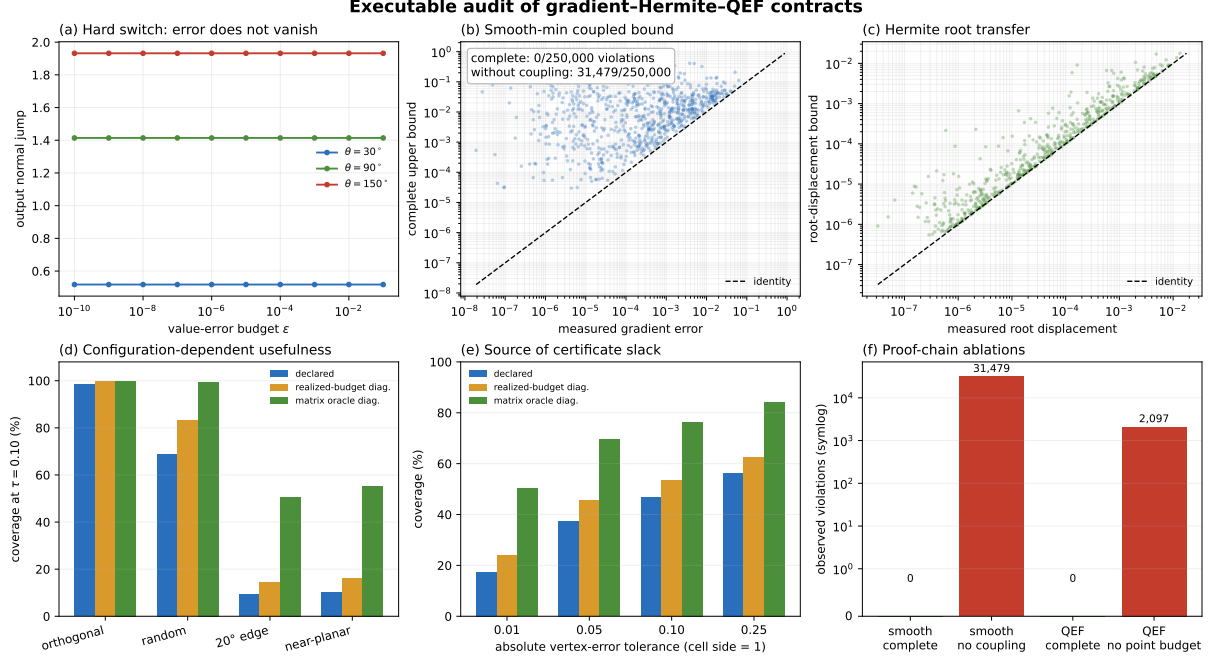


Figure 1: Executable audit. (a) The hard-switch normal jump remains fixed as the input value error tends to zero. (b,c) Complete smooth-min and edge-root bounds dominate measured errors. (d) At $\tau = 0.10$, declared-contract coverage is much lower on edge and near-planar configurations; realized-budget and exact-matrix diagnostics expose where slack enters. (e) The same three arms are compared across tolerances. (f) Complete bounds show zero observed violations, while deleting the coupling or Hermite point budget yields counterexamples. Finite sampling validates the implementation, not the universal theorems.

A finite worst-case bound need not be useful. The median QEF bound is 0.1343 cell units, whereas the median measured error is 0.000592; the median bound-to-error ratio is 484. Requiring bounds of at most 0.01, 0.05, 0.10, and 0.25 cell units yields total-query coverage of 17.42%, 37.38%, 46.75%, and 56.34%. This conservatism is a central result rather than a detail to hide. It motivates tighter structured perturbation bounds, adaptive subdivision, and a cost-coverage analysis in the full system.

The diagnostics localize part of that conservatism. Across all configurations, the median declared, realized-budget, and exact-matrix bounds are 0.13431, 0.07074, and 0.00987 cell units. At $\tau = 0.10$, their respective coverage is 46.75%, 53.52%, and 76.32%. The gap between the first two columns is attributable to unused width in the declared input contracts; the remaining gap shows that row aggregation, triangle inequalities, and the observed-side spectral lower bound also matter. In the deployable arm, coverage at this tolerance is only 9.33% on the 20° edge configuration and 10.38% on near-planar configurations. The current certificate therefore does not provide useful coverage for most such cells at this tolerance, even though its finite bounds remain sound in the audit.

Remark 15 (Source of the observed coverage gap). At $\tau = 0.10$, measured QEF displacement is within tolerance in 39,939 of the 40,000 conditioning trials (99.8475%), whereas the deployable certificate accepts 18,700 (46.75%). Thus 53.0975 percentage points are audit false negatives attributable to certificate slack, while 0.1525 percentage points correspond to measured displacements that actually exceed the tolerance. In this audit, approximately 99.71% of the non-certification gap is therefore associated with conservatism rather than an above-tolerance measured displacement. At the level of global medians, the declared-to-realized and realized-to-matrix ratios are 1.90 and 7.17. These are descriptive ratios, not a causal decomposition, but

Table 2: Configuration-wise QEF usefulness and diagnostic decomposition. “Realized” uses the errors that occurred in each trial; “matrix” uses exact matrix perturbations and the true spectral denominator. Both are nondeployable diagnostics. Coverage uses $\tau = 0.10$ cell units.

Configuration	data ID	median actual	declared	realized	matrix	declared coverage
Orthogonal	100.00%	0.000186	0.01198	0.00653	0.00139	98.31%
Random	99.11%	0.000245	0.04317	0.02324	0.00397	68.98%
20° edge	0.00%	0.000972	1.60751	0.89061	0.09671	9.33%
Near planar	4.85%	0.001611	1.54401	0.82798	0.07484	10.38%

Table 3: Residual-stress extension of the QEF audit. “Measured” and “declared” are coverage at $\tau = 0.10$. The compatible baseline is the original 40,000-system conditioning audit.

System class	tested	median residual	median actual	median bound	measured	declared	failures
Compatible common point	40,000	1.25×10^{-16}	0.000592	0.13431	99.8475%	46.75%	0
Inconsistent analytic planes	10,000	0.00295	0.000626	0.11977	99.73%	47.98%	0
Sphere/paraboloid patches	10,000	0.02348	0.001339	0.37076	99.99%	27.86%	0

they indicate that analytical aggregation is the larger source of slack in the present experiment.

The residual-stress audit tests whether soundness depended on the compatible common-point construction. Its 20,000 complete bounds have zero observed violations, with a one-sided 95% all-success lower bound of 99.9850%. The median minimum unregularized data residual rises from numerical zero in the compatible baseline to 0.00295 for inconsistent analytic planes and 0.02348 for curved Hermite patches (table 3 and fig. 2). The complete bound remains sound, whereas deleting the Hermite point budget causes 777 violations in these added systems.

The ablations demonstrate that the complete chain is doing logical work. Removing the smooth value–gradient coupling produces 31,479 failures among 250,000 tests (12.59%). Keeping the QEF spectral guard but omitting the Hermite point-position budget produces 2,097 bound violations among the 40,000 compatible systems and another 777 among the 20,000 residual-stress systems. Thus normal uncertainty alone cannot certify the QEF right-hand side.

9 Real-CAD end-to-end audit

We additionally audited a frozen corpus of 528 STEP files: 28 manually curated public or industrial models and a deterministic 500-model Fusion 360 Gallery sample. OpenCascade 7.9.3 imported all files. The complete imported shapes and the extracted solid subsets passed the kernel validity check, and every solid subset had closed shells and positive B-rep volume. Without healing, sewing, or fusion, the derived-mesh gate admitted 511 models; eight meshes had boundary edges and nine had non-manifold edges. All failures remain in the denominator. The frozen 50-to-28 manual-candidate ledger records all 33 STEP members of the NIST archive and all 17 external download receipts, together with each inclusion or exclusion reason. Before verification, the archive and all 528 selected STEP inputs are independently rehashed against their receipts or deterministic Fusion sample manifest.

An isolated FreeCAD 1.1.1 pipeline backed by OpenCascade 7.8.1 also imported all 528 files. The two pipelines agreed on validity, solid/shell/face counts, auxiliary-shell inventory, closure, and positive-volume classification; every B-rep volume pair agreed within 10^{-4} relative error (maximum 5.99e-05). This is a cross-version and cross-frontend consistency check within the OpenCascade family, not an independent second-kernel result.

The end-to-end arm used all 22 admitted manual models and the first 100 admitted Fusion rows in frozen sample order. Manual and Fusion models used respectively 32 and 24 cells along the longest box axis. Each isolated run performed STEP import, solid-subset tessellation,

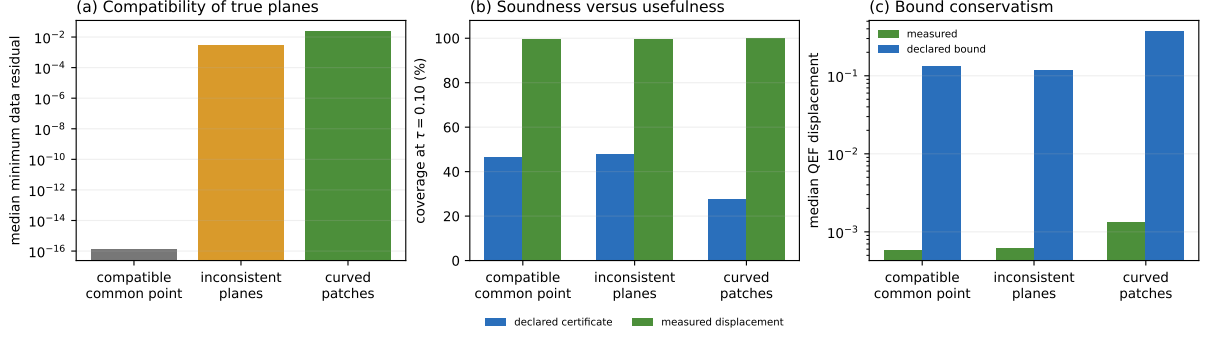


Figure 2: Audit beyond common-point Hermite systems. (a) Inconsistent analytic planes and curved patches have nonzero minimum data residuals. (b) Complete bounds remain sound, but declared coverage remains much lower than measured within-tolerance displacement. (c) The declared bounds remain conservative. Finite sampling checks the implementation and does not prove the theorem.

Table 4: No-healing preflight admission by source. Boundary and non-manifold columns are mutually exclusive rejection reasons.

Source	Total	Admitted	Boundary	Non-manifold
3D ContentCentral	4	4	0	0
3Dfindit	4	1	0	3
Fusion360Gallery	500	489	8	3
GrabCAD	5	4	0	1
NIST	11	11	0	0
TraceParts	4	2	0	2

signed-distance sampling, Hermite extraction, regularized per-cell QEF solution ($\lambda = 0.1$), and uniform-grid Dual Contouring connectivity.

The main run completed 118/122 models. Four thin or anisotropic Fusion parts had no grid-node sign change at resolution 24; a diagnostic-only ladder recovered two at resolution 32 and two at resolution 48, without replacing the four main failures. Across 162,216 active cells, 36,866 met the $\tau = 0.10$ oracle-diagnostic bound (22.73% pooled coverage), and the complete displacement bound had zero violations. The model-wise median coverage was 13.76%. The model-median normal-angle error was 3.72 degrees, while the 95th percentile across model-level 95th-percentile errors was 74.29 degrees.

The extracted meshes had zero boundary edges and zero degenerate triangles, but 2,632 non-manifold edges across 79/118 completed models. This exposes a connectivity limitation of standard sign-pattern Dual Contouring rather than a contract failure, and motivates manifold or adaptive connectivity in future work. The median model mean surface error was 0.0462 cell units.

The real-CAD files do not provide certified value and gradient budgets. We therefore use source-mesh closest points and pseudo-normals only to audit the theorem-to-implementation chain. These measured errors are nondeployable and must not be labelled CERTIFIED. An explicit cell-centre fallback for UNKNOWNcells increased the median mean error from 0.0462 to 0.2842 cell units, confirming that the contract is a status and admission mechanism rather than an automatic geometry repair.

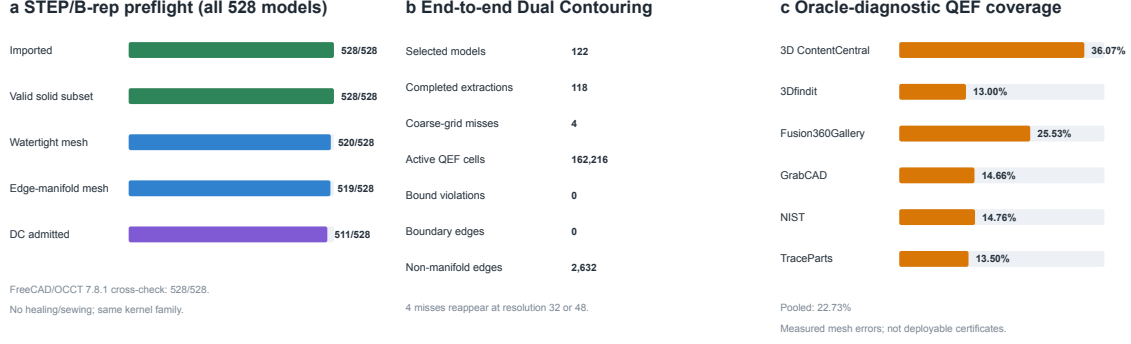


Figure 3: Real-CAD audit. (a) The no-healing STEP/B-rep preflight retains all 528 inputs in the denominator and admits 511 derived meshes. (b) The fixed regular-grid run completes 118 of 122 selected models; the four coarse-grid misses reappear at resolution 32 or 48 in a diagnostic-only ladder. (c) Oracle-diagnostic QEF coverage varies by source. Because the source mesh provides the reference closest point and pseudo-normal, these coverage values are not deployable certificates.

Table 5: Real-CAD end-to-end results. Coverage uses the source mesh as an oracle and is not a deployable certificate.

Source	Selected	Completed	Active cells	Coverage
3D ContentCentral	4	4	11,522	36.07%
3Dfindit	1	1	7,037	13.00%
Fusion360Gallery	100	96	99,516	25.53%
GrabCAD	4	4	15,125	14.66%
NIST	11	11	20,387	14.76%
TraceParts	2	2	8,629	13.50%

10 Discussion

Single normals versus sharp features. The negative result is not an argument for smoothing every Boolean crease. At a switch, the two active normals contain useful feature information. The contract should preserve them as separate Hermite constraints or a normal set. It returns UNKNOWN only when a downstream interface insists on one normal that the upstream geometry does not uniquely define.

Worst case versus probabilistic uncertainty. The QEF bound is conservative because it assumes adversarial alignment of all plane perturbations. Probabilistic quadrics can be substantially more useful when a credible distribution is available [2]. The two approaches answer different questions. A promising hybrid would use the deterministic contract as a safety envelope and a probabilistic model to choose a high-quality point within it.

Regularization and bias. Regularization makes the QEF strongly convex and simplifies certification, but it changes the target minimizer. Our theorem compares approximate and true solutions at the same λ ; it does not bound bias relative to the unregularized surface-feature intersection. Choosing λ therefore requires a separate accuracy-conditioning tradeoff.

Relation to adaptive Dual Contouring. The current result certifies one cell’s inputs and regularized QEF solution. A complete adaptive algorithm still needs certified active-cell detection,

consistent decisions across levels, topology or manifold conditions, and a crack-free connectivity theorem. The real-CAD audit implements a separate uniform-grid Dual Contouring pipeline and exposes non-manifold connectivity on 79 of 118 completed models. The current HybridCADLab production path still uses Marching Cubes; the audit should therefore not be read as an adaptive or production Dual Contouring backend. Existing manifold and isotopy-preserving methods [3, 4] clarify the missing global obligations, but do not eliminate the local input-uncertainty problem addressed by the present contract.

11 Limitations and next steps

First, the contract consumes value and gradient budgets. Analytic primitives can supply them, but sampled grids and neural fields need a separate producer, possibly based on interval interpolation, verified automatic differentiation, or Lipschitz-constrained networks. Empirical residuals are not automatically valid worst-case budgets.

Second, the formal hard-Boolean result covers branch separation and a local set-valued switch enclosure. A deep DAG theorem must propagate validity domains and active sets through union, intersection, and difference without silently turning a local certificate into a global claim.

Third, the binary smooth-min bound is closed, but an (n)-ary result should use the full softmax Jacobian rather than repeatedly applying a binary bound whose tightness depends on tree association.

Fourth, the real-CAD arm closes the previous end-to-end implementation gap but does not close the producer-certificate gap. It uses a source triangle mesh to obtain reference closest points and pseudo-normals, so its coverage is an offline oracle diagnostic. The uniform grid also misses four thin or anisotropic parts at the frozen coarse resolution, and standard sign-pattern connectivity produces non-manifold edges on many completed models. The next engineering stage is an adaptive, manifold-aware contourer driven by an actual producer of certified value and gradient budgets. A larger corpus run is justified only after those upstream certificates execute rather than replaying metadata or substituting empirical residuals.

Finally, the present bounds are often loose. The current row-wise budgets enter through the Frobenius aggregation η_N . A tighter deployable bound cannot be justified from the smaller operator norm observed against an unknown true matrix; it requires an upstream operator-norm or correlation-aware certificate as in remark 11. Such structured perturbation budgets, interval eigenvalue bounds, adaptive cell subdivision, and normal-cone-aware QEF formulations are the main routes to improve useful coverage without weakening soundness.

12 Conclusion

A value-error budget is not a normal certificate, and a normal certificate is not a QEF certificate. The missing links are branch state, root position, normal normalization, local plane offset, and matrix conditioning. We have made these links explicit in a fail-closed cell-level contract. A hard Boolean switch requires either certified branch separation or a set-valued normal conclusion. A smooth Boolean requires a value-gradient coupling term. A QEF requires both normal and point budgets and an observed-side spectral guard. The analytic stress tests support the implementation and falsify two tempting simplifications, while also showing that worst-case QEF bounds can be highly conservative. The added non-concurrent and curved-patch systems show that the zero-violation result is not restricted to compatible, common-point planes. The 528-model STEP preflight and 122-model regular-grid contouring arm further show that the implementation runs on heterogeneous CAD geometry, while its 22.73% oracle-diagnostic coverage and non-manifold outputs expose the remaining producer and connectivity gaps. Completing a production adaptive contourer is future work; the current contribution is the theorem and software interface that such a system must satisfy before it labels a Hermite/QEF vertex as certified.

Reproducibility statement

The research package contains the synthetic generator, frozen JSON record, regression checks, figure scripts, proof-status and audit records, and this manuscript source. It also contains the real-CAD import and contouring source, input receipts and hashes, row-level results, cross-version comparison, 50-to-28 selection ledger, and a fail-closed verifier with negative tamper tests. Third-party STEP files are not redistributed; their source records and cryptographic hashes are provided. The synthetic generator uses fixed seed 20260808. Figure code reads frozen results and does not rerun experiments. A success message explicitly states that finite sampling is not a proof.

Conflict of interest

The author declares no conflict of interest.

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Data availability

The source code, fixed seeds, frozen numerical results, row-level real-CAD records, regression checks, input receipts, verification scripts, and figure-generation scripts supporting the findings of this study are provided as Supporting Information and in the public artifact repository <https://github.com/Jeremy-Pei/gradient-qef-contracts-artifact>. Third-party STEP files are not redistributed; the repository provides acquisition records and SHA-256 values so that authorized users can reconstruct and verify the input corpus.

Ethics statement

Ethics approval, consent to participate, and consent for publication are not applicable because this mathematical and computational study involved no human participants, animals, or identifiable personal data.

CRedit authorship contribution statement

Jeremy Pei: Conceptualization; Methodology; Formal analysis; Software; Investigation; Validation; Visualization; Writing – original draft; Writing – review and editing.

Declaration of generative AI and AI-assisted technologies

During preparation of this manuscript, the author used OpenAI Codex to help organize the research plan, draft and inspect mathematical arguments, implement analytic stress tests, check reproducibility, and edit the language. The author independently reviewed and revised the mathematical arguments, references, source code, numerical results, and wording, and takes full responsibility for the submitted content.

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